



Source Identification for Discrete Mixtures of Product Distributions

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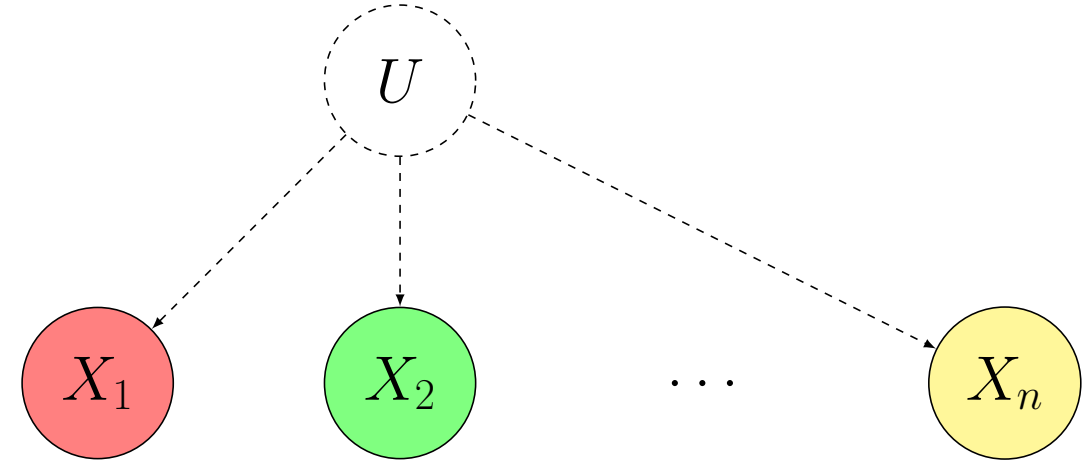
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The Problem

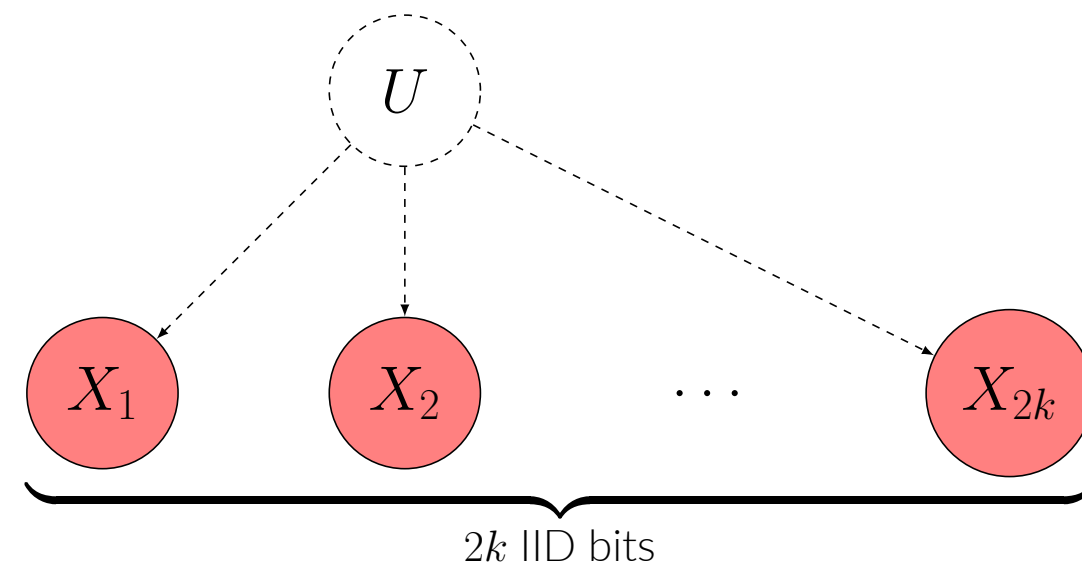


- Source U has k states: $\pi_j := \Pr(U = j)$. bits $X_1, \dots, X_n$ are independent conditioned on U
- The distribution is parameterized by a matrix $\mathbf{m} \in \mathbb{R}^{n \times k}$ where
$$\mathbf{m}_{ij} := \Pr(X_i = 1 \mid U = j)$$
- The *goal* is to reconstruct the parameters $\pi, \mathbf{m}_{ij}$ (up to an additive ε)
- Strategy:** Breaking of independence properties between variables gives information about U .

Background Literature and Contribution

- Learning: Finding a model that generates a distribution within statistical ε of the true distribution
 - (Feldman, O'Donnell, and Servedio, 2008) gives an $(nk/\varepsilon)^{O(k^3)}$ time algorithm
 - (Chen and Moitra, 2019) improved this to $k^{O(k^3)}(n/\varepsilon)^{O(k^2)}$.
- Identification: Finding a model that is close in parameter space
 - Separation* of parameters as well as nontrivial mixture weights become important.
 - (Tahmasebi, Motahari, and Maddah-Ali, 2018) showed having $2k - 1$ observable variables that are separated is sufficient, but did not focus the algorithm itself.
- We assume that enough bits satisfy ζ -separation: $\mathbf{m}_{ij} - \mathbf{m}_{ij'} \geq \zeta$.
- We achieve accuracy of $\zeta^{-O(k^2 \log k)} \varepsilon$ with $2^{O(k^2)} n^{O(k)}$ arithmetic operations
- Previous $k^{O(k^3)}(n/\varepsilon)^{O(k^2)}$ for algorithm that does not *identify* the source
- We accomplish all of this by reducing to the ``Power Distribution" case.

Power Distributions



- Equivalent to choosing a coin with $\Pr(\text{Heads}) = \mathbf{m}_{ij}$ with probability π_j and flipping it $2k$ times.
- Data may look like the following table:

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
Sample 1	0	1	0	0	0	0	0	0
Sample 2	1	1	1	0	1	1	1	0
Sample 3	1	0	1	1	1	0	1	1
Sample 4	0	0	1	0	0	1	0	0

Power Distribution Moments

Power Distribution Moments

For random variables X_i as above $X_i^{\odot 2}$ is the random variable obtaining by taking the product of X_i and an identically distributed copy of X_i that is independent given U .

- In this special case, $\mathbf{m}_{ij} = \mathbf{m}_{1j} =: \alpha_j$ for all $i \in [n]$, we can look at *moments*:

$$\mu_\ell = \mathbb{E}[X_1^{\odot \ell}] = \mathbb{E}[X_1 X_2 X_3 \cdots X_\ell] = \sum_j \pi_j \alpha_j^\ell$$

- .
- We can estimate moments $\mu_0, \mu_1, \mu_2, \dots$, etc from samples via counting.

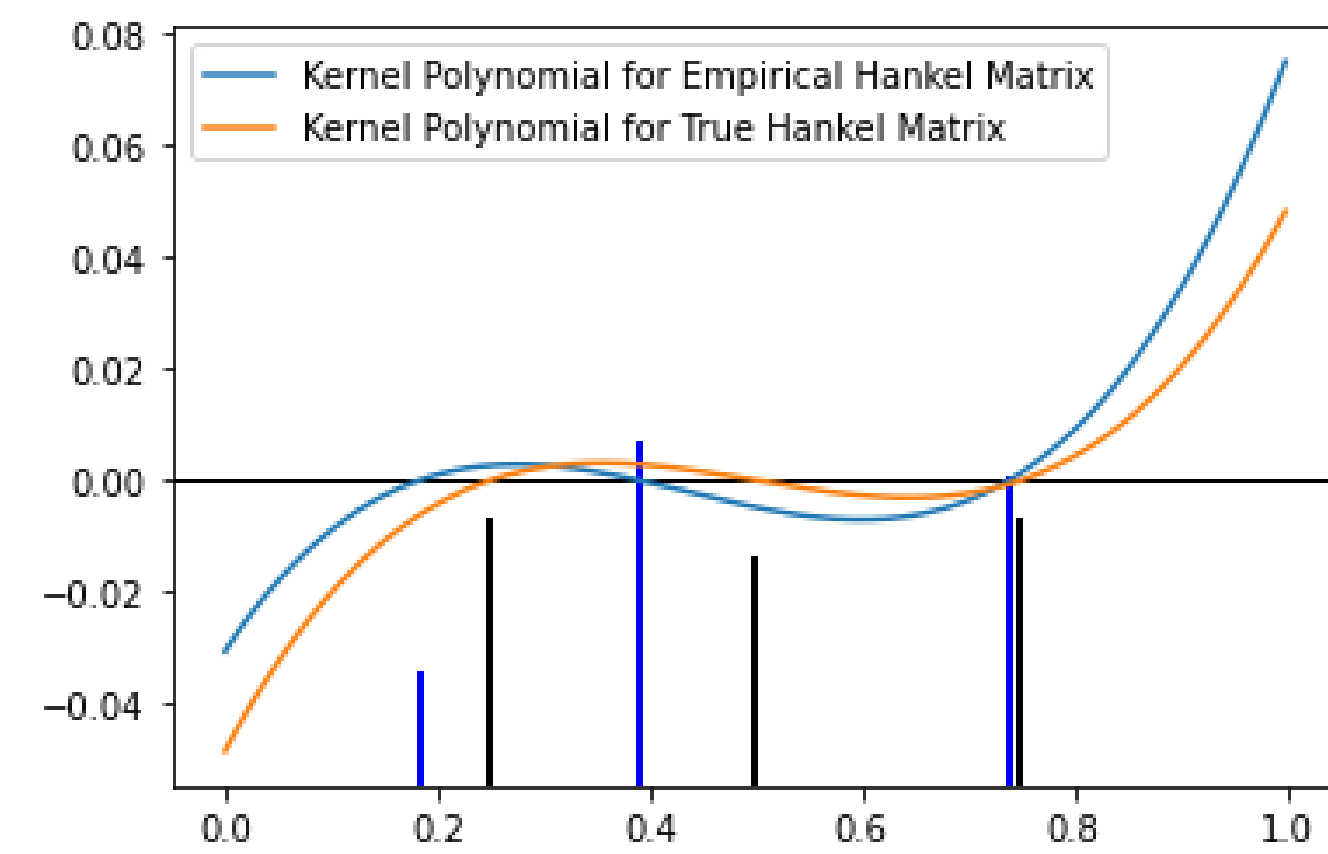
Hankel Matrix

- The Hankel matrix for the distribution $\Pr(X)$ is

$$\mathcal{H}_{k+1} = \begin{bmatrix} \mu_0 & \mu_1 & \mu_2 & \cdots & \mu_k \\ \mu_1 & \mu_2 & \mu_3 & \cdots & \mu_{k+1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mu_k & \mu_{k+1} & \mu_{k+2} & \cdots & \mu_{2k} \end{bmatrix},$$

- The empirical Hankel matrix $\tilde{\mathcal{H}}_{k+1}$ has estimates of the moments as entries.

Prony's Method



- The kernel of $\tilde{\mathcal{H}}_{k+1}$ has dimension 1 and is spanned by a vector $c = (c_0, \dots, c_k)$.
- Define the *kernel polynomial* $p(x) := c_0 + c_1 x + \dots + c_k x^k$.
- The roots of p are close to $\{\alpha_1, \dots, \alpha_k\}$.

To solve for α, π :

- Find the kernel of $\tilde{\mathcal{H}}_{k+1}$, c
- Find the roots $\{\alpha_1, \dots, \alpha_k\}$ of the corresponding p .
- Solve for π in

$$\begin{bmatrix} \mu_0 \\ \mu_1 \\ \vdots \\ \mu_{k-1} \end{bmatrix} = \begin{bmatrix} \alpha_1^0 & \alpha_2^0 & \alpha_3^0 & \cdots & \alpha_k^0 \\ \alpha_1^1 & \alpha_2^1 & \alpha_3^1 & \cdots & \alpha_k^1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha_1^{k-1} & \alpha_2^{k-1} & \alpha_3^{k-1} & \cdots & \alpha_k^{k-1} \end{bmatrix} \begin{bmatrix} \pi_1 \\ \pi_2 \\ \vdots \\ \pi_k \end{bmatrix}$$

Details and error analysis in (Gordon et al., 2020).

Hadamard Products and Synthetic Bits Preliminaries

Hadamard products

The *Hadamard product* between $x, y \in \mathbb{R}^{m \times n}$ is $x \odot y \in \mathbb{R}^{m \times n}$, defined by $(x \odot y)_{ij} = x_{ij} y_{ij}$.

- Let $\mathbf{M} \in \mathbb{R}^{2^{[n]} \times k}$ be the matrix with rows indexed by $S \subseteq [n]$ where

$$\mathbf{M}_S := \mathbf{m}_{i_1} \odot \mathbf{m}_{i_2} \odot \cdots \odot \mathbf{m}_{i_r}.$$

- In particular, $\mathbf{M}_\emptyset = (1, 1, \dots, 1)$ and $\mathbf{M}_{\{i\}} = \mathbf{m}_i$ for all $i \in [m]$. We define $X_S := \prod_{s \in S} X_s$ for $S \subseteq [n]$ so that $\mathbb{E}[X_S] = \mathbf{M}_S \pi$.
- For a given subset $\mathcal{S} \subseteq 2^{[n]}$, define $\mathbf{M}[\mathcal{S}]$ to be the matrix with rows $\mathbf{M}_S$ for each $S \in \mathcal{S}$.
- Given disjointly supported $\mathcal{S}, \mathcal{T}$, define $\mathbf{C}_{\mathcal{S}, \mathcal{T}} := \mathbf{M}[\mathcal{T}] \text{diag}(\pi) \mathbf{M}[\mathcal{S}]^\top$.
- We can estimate all entries of $\mathbf{C}_{\mathcal{S}, \mathcal{T}}$!

Theorem

Assuming ζ -separation for at least $2k$ bits, there exist disjointly-supported sets of k rows each, $\mathcal{A} = \{A_1, \dots, A_k\}, \mathcal{B} \subseteq 2^{[n] \setminus \{1\}}$, such that $\mathbf{M}[\mathcal{B}], \mathbf{M}[\mathcal{A}]$ are full-rank and each have condition number $O(\zeta^{-k^2})$! (We require that $A_1 = \emptyset$.)

So, the product $\mathbf{C}_{\mathcal{B}, \mathcal{A}} := \mathbf{M}[\mathcal{B}] \text{diag}(\pi) \mathbf{M}[\mathcal{A}]^\top$ is also invertible with bounded condition number and each entry can be estimated empirically from samples.

Synthetic Bits

- The value $\mathbb{E}[X_1]$ is observable. We want to compute $\mathbb{E}[X_1^{\odot 2}], \mathbb{E}[X_1^{\odot 3}], \dots$ to be able to use the algorithm for the power distribution!
- Instead we compute

$$\begin{aligned} v_r &:= (\mathbb{E}[X_1^{\odot r} X_{A_1}], \mathbb{E}[X_1^{\odot r} X_{A_2}], \dots, \mathbb{E}[X_1^{\odot r} X_{A_k}]) \\ &= \mathbf{m}_1^{\odot r} \text{diag}(\pi) \mathbf{M}[\mathcal{A}]^\top \end{aligned}$$

for $r = 1, \dots, 2k$.

- Since $A_1 = \emptyset$,

$$v_r = (\mathbb{E}[X_1^{\odot r}], \mathbb{E}[X_1^{\odot r} X_{A_2}], \dots, \mathbb{E}[X_1^{\odot r} X_{A_k}])$$

- We can estimate v_1 directly:

$$\begin{aligned} v_1 &= (\mathbb{E}[X_1 X_{A_1}], \mathbb{E}[X_1 X_{A_2}], \dots, \mathbb{E}[X_1 X_{A_k}]) \\ &= (\mathbb{E}[X_{A_1 + \{1\}}], \mathbb{E}[X_{A_2 + \{1\}}], \dots, \mathbb{E}[X_{A_k + \{1\}}]) \end{aligned}$$

- To compute v_2 , we will synthesize an independent copy of X_1 from the rows in $\mathcal{B}$!
- Suppose we had a $u_1 \in \mathbb{R}^k$ such that $u_1 \mathbf{M}[\mathcal{B}] = \mathbf{m}_1$.
- Then $u_1 \mathbf{M}[\mathcal{B} + \{1\}] = \mathbf{m}_1^{\odot 2}$
- Since $\text{diag}(\pi) \mathbf{M}[\mathcal{A}]^\top$ is invertible,

$$\begin{aligned} u_1 \mathbf{M}[\mathcal{B}] = \mathbf{m}_1 &\iff u_1 \underbrace{\mathbf{M}[\mathcal{B}] \text{diag}(\pi) \mathbf{M}[\mathcal{A}]^\top}_{\mathbf{C}_{\mathcal{B}, \mathcal{A}}} = \underbrace{\mathbf{m}_1 \text{diag}(\pi) \mathbf{M}[\mathcal{A}]^\top}_{v_1} \\ &\iff u_1 \mathbf{C}_{\mathcal{B}, \mathcal{A}} = v_1 \iff u_1 = v_1 \mathbf{C}_{\mathcal{B}, \mathcal{A}}^{-1} \end{aligned}$$

- Now we multiply by $\mathbf{C}_{\mathcal{B} + \{1\}, \mathcal{A}}$ to get

$$\begin{aligned} u_1 \mathbf{C}_{\mathcal{B} + \{1\}, \mathcal{A}} &= u_1 \mathbf{M}[\mathcal{B} + \{1\}] \text{diag}(\pi) \mathbf{M}[\mathcal{A}]^\top \\ &= \mathbf{m}_1^{\odot 2} \text{diag}(\pi) \mathbf{M}[\mathcal{A}]^\top \\ &= (\mathbb{E}[X_1^{\odot 2} X_{A_1}], \mathbb{E}[X_1^{\odot 2} X_{A_2}], \dots, \mathbb{E}[X_1^{\odot 2} X_{A_k}]) = v_2 \end{aligned}$$

- Since no subset in $\mathcal{B}$ uses X_1 , we can estimate $\mathbf{C}_{\mathcal{B} + \{1\}, \mathcal{A}}$ from samples!

Overview of Synthetic Bits

$$\begin{aligned} \text{Definition} \quad v_r &:= \mathbf{m}_1^{\odot r} \pi_\odot \mathbf{A}^\top &= (\mathbf{m}_1 \odot (u_{r-1} \mathbf{B})) \pi_\odot \mathbf{A}^\top &= u_{r-1} \mathbf{C}_{\mathcal{B} + \{1\}, \mathcal{A}} &(2) \\ u_r &:= \mathbf{m}_1^{\odot r} \mathbf{B}^{-1} &= \mathbf{m}_1^{\odot r} \pi_\odot \mathbf{A}^\top \mathbf{A}^{\top^{-1}} \pi_\odot^{-1} \mathbf{B}^{-1} &= v_r \mathbf{C}_{\mathcal{B}, \mathcal{A}}^{-1} &(3) \end{aligned}$$

The Rest of the Algorithm

- By repeatedly computing u_r and v_r for $r \leq 2k$, we obtain approximate moments $\mathbb{E}[X_1^{\odot r}]$ for $r = 1, 2, \dots, 2k$; we can then use Prony's method to solve for $\mathbf{m}_1$ and π ; finally, we solve for $\mathbf{m}$.
- By bounding the condition number of $\mathbf{C}_{\mathcal{B} + \{1\}, \mathcal{A}}$ and $\mathbf{C}_{\mathcal{B}, \mathcal{A}}$ we bound the error introduced in each iteration.
- Each iteration increases the error by $O(\zeta^{-k^2})$.
- If we assume ζ -separation for at least $3k$ -bits and use disjointly-supported sets $\mathcal{B}, \mathcal{B}', \mathcal{A}$, and corresponding matrices $\mathbf{C}_{\mathcal{B}, \mathcal{A}}, \mathbf{C}_{\mathcal{B}', \mathcal{A}}$, and $\mathbf{C}_{\mathcal{B} + \mathcal{B}', \mathcal{A}}$, we can get away with only $\log k$ iterations to compute $\mathbb{E}[X_1^{\odot k}]$.

Open Questions

- Open question:** Is this condition number bound tight?
- For the power distribution case (where $\mathbf{m}$ is a Vandermonde matrix), the condition number is $O(\zeta^{-k})$.
- An improvement to $O(\zeta^{-k})$ would immediately imply that our algorithm would only increase error by a factor of $O(\zeta^{-k \log k})$.

References

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